EE5203 L11 Study Sheet

I2C and SPI - Basri Erdogan

Essential question: A UART assumes a rate, reaches one place, and never confirms anything. Add a clock wire and all three change. What exactly do you gain, and what do you pay?

What you should be able to do

- Explain what a clock line buys over the UART's assumed rate.
- Read an I2C transaction: START, address, R/W, ACK, data, STOP.
- Convert between 7-bit and HAL's 8-bit address form, both directions.
- Explain open drain and why pull-ups are mandatory.
- State CPOL/CPHA from a timing diagram and name the mode.
- Choose a bus for a stated device and pin budget.

Synchronous versus asynchronous

	UART (L10)	I2C	SPI
Clock line	none	`SCL`	`SCK`
Wires	2	2 + pull-ups	3 + one CS each
Addressing	none	7-bit, in protocol	physical (CS)
Acknowledgement	none	yes (ACK/NACK)	none
Typical speed	9.6–115 kbaud	100–400 kHz	1–50 MHz
Detects missing device	no	**yes**	no

The clock removes the rate assumption; the address removes the one-to-one restriction; the ACK removes the silence. SPI keeps the clock but drops the other two in exchange for speed.

I2C: the transaction

```
START | 7-bit addr + R/W | ACK | data byte | ACK | ... | STOP
  |         |               |                 |            |
 master    master        slave    master     slave      master
```

- Both lines are **open drain**: a device can only pull *low*. External **pull-ups** (4.7 kΩ) return them high. No pull-ups → lines stuck low → every transfer times out.
- Only the addressed device responds; all the others ignore the traffic.
- **ACK** = receiver pulled `SDA` low for one clock. **NACK** = left high.

The address shift

Form	Value	Used by
7-bit	`0x27`	datasheets, bus scanners, humans
8-bit	`0x4E` = `0x27 << 1`	**STM32 HAL**

```
HAL_I2C_Master_Transmit(&hi2c1, (0x27 << 1), buf, n, 100);   /* or 0x4E */
```

Passing `0x27` directly addresses device `0x13`, which is not there — the single most expensive mistake on this bus. If a device will not answer, check the shift **before** the wiring.

I2C timing

Mode	Clock	Bit	Byte + ACK (9 bits)
Standard	100 kHz	10 µs	90 µs
Fast	400 kHz	2.5 µs	22.5 µs

The LCD backpack is a **port expander**: one character = **four** I2C bytes (two nibbles, each strobed with the enable line). So one character 45 bits **450 µs** at 100 kHz, and a 16-character line **7.2 ms** — over seven tick periods. Refresh rate is a design decision with a measurable cost.

SPI: the four wires

Wire	Meaning
`SCK`	clock, from the master
`MOSI`	master out, slave in
`MISO`	master in, slave out
`CS`	chip select, active low, **one per device**

Every transfer is a **full-duplex exchange**: for each bit sent on `MOSI`, one arrives on `MISO`. To read, you must still write — usually a dummy byte.

```
HAL_GPIO_WritePin(CS_PORT, CS_PIN, GPIO_PIN_RESET);   /* select   */
HAL_SPI_TransmitReceive(&hspi2, tx, rx, n, 100);
HAL_GPIO_WritePin(CS_PORT, CS_PIN, GPIO_PIN_SET);     /* deselect */
```

`CS` is an ordinary GPIO — HAL does not drive it. It must stay low for the whole transaction; releasing it between bytes resets most slaves.

CPOL, CPHA, and the four modes

Mode	CPOL	CPHA	Clock idles	Sampled on
0	0	0	low	leading (rising) edge
1	0	1	low	trailing (falling) edge
2	1	0	high	leading (falling) edge
3	1	1	high	trailing (rising) edge

Mode 0 is the most common. The **slave's** datasheet decides; the master obeys. A wrong mode does not give silence — it gives consistent, bit-shifted garbage, which is much harder to recognise.

Loopback

One jumper from `MOSI` to `MISO` makes the bus talk to itself:

```
uint8_t tx = 0xA5, rx = 0;
HAL_SPI_TransmitReceive(&hspi2, &tx, &rx, 1, 100);   /* rx == 0xA5 */
```

This separates "my SPI is misconfigured" from "the slave is not answering," and needs no slave. Remove the jumper and `rx` becomes `0x00` or `0xFF` with **no error** — SPI cannot tell you a device is missing.

SFR evidence

Claim	Evidence
the I2C pins are open drain	`GPIOB->OTYPER` bits 8, 9 = 1
nothing answered	`I2C1->SR1` `AF` (acknowledge failure) = 1
the bus is configured as I expect	`I2C1->CCR` for the clock rate
my SPI mode is what the slave wants	`SPI2->CR1` `CPOL`/`CPHA` bits
the SPI link physically works	loopback: `SPI2->DR` returns what you wrote

Common mistakes

- The 7-bit/8-bit shift — `0x27` passed where `0x4E` is wanted.
- No pull-ups: lines low, everything times out.
- `SDA`/`SCL` swapped — a bus scan then finds nothing at all.
- Ignoring the HAL return value; a NACK is real information.

- Wrong SPI mode: plausible but shifted data.
- `CS` released between bytes of one transaction.
- Expecting SPI to report a missing slave. It cannot.
- Refreshing the whole LCD every tick — 7.2 ms of I2C per line.

Mastery self-check

- ☐ I can name the three things a clock line and addressing buy over a UART.
- ☐ I can label START, address, R/W, ACK, and STOP on a trace.
- ☐ I can convert 7-bit 8-bit addresses in both directions.
- ☐ I can explain open drain and predict the failure without pull-ups.
- ☐ I can compute I2C byte and line times at 100 and 400 kHz.
- ☐ I can derive CPOL/CPHA from a timing diagram and name the mode.
- ☐ I can explain what loopback proves and what it does not.
- ☐ I can justify a bus choice from device count and pin budget.

Five review questions

1. A datasheet gives an address of `0x68`. Give the value for `HAL_I2C_Mem_Read`, and explain what would be addressed if you passed `0x68`.
2. Your bus scan finds nothing at all. List three distinct causes and the test that distinguishes each.
3. A 16-character line costs 7.2 ms at 100 kHz. You want the display and a 1 ms tick to coexist. Choose a refresh rate and defend it with numbers.
4. A slave's timing diagram shows the clock idling low and data sampled on the falling edge. Give CPOL, CPHA, and the mode. What do you see if you configure mode 0?
5. You must add eight identical sensors that each report once per second, and you have four spare pins. Choose a bus and justify it — then say what would change your answer.

See the L11 worksheet for extended practice. Worked solutions are released after the practice window.